

EE 101: Electrical Science

Course Code	EE 101
Course Name	Electrical Science
L-T-P-C	3-1-2-4
Prerequisites	NIL Sem. Both
Approval	5th Senate (Oct 2013)

Equivalent Course: IC 160

Course Contents

- **Network Fundamentals:** Types of Sources and Elements, Kirchoff's Laws, Mesh and Node Analysis of D.C. Networks.
- **Transient Analysis:** RL & RC circuits.
- **Network Theorems:** Thevenin's Theorem, Norton's Theorem, Superposition Theorem, Maximum Power Theorem, Star-Delta Transformation.
- **A.C. Fundamentals:** Concept of Phasor, Impedance and Admittance, Mesh and Node analysis of Single Phase AC Networks, Network Theorems in AC Networks, Active and Reactive Power in AC Circuits, Resonance in Series AC Circuits.
- **Introduction to 3-phase A.C. Circuits:** Analysis of 3-phase balanced star-delta circuits, Power in 3-phase Circuits.
- **Magnetic Circuit Concepts:** Analogy with Electrical Circuits, Calculation for series, parallel and series parallel magnetic circuits, Eddy current and Hysteresis losses.
- **Single Phase Transformer:** Basic constructional features, Operating principle, Phasor diagram, Equivalent Circuit, Voltage regulation, Efficiency, Open circuit and Short Circuit tests.
- **D.C. Machines :** Principle of operation, Basic constructional features, Emf and torque equation, Armature reaction, Types of Excitation and Generator characteristics, Types of D.C. motors, Starting and speed control of D.C. motors.
- **Machines:** Three phase Induction Motor: Operating principle, Constructional features, Equivalent circuits, Torque-speed characteristics, Starting and speed control. Synchronous Generator: Basic principle of operation, Emf equation, Constructional features.
- **Measurement of Electrical Quantities:** Measurement of Voltage, Current, Power and Energy, Moving Iron Instruments, Measurement of 3 phase power, Accuracy class of meters.

References

1. Mukhopadhyaya P., Pant A. K., Kumar V. and Chittore D. S., **Elements of Electrical Science**, M/s Nem Chand & Brothers.
2. Vincent Del Toro, **Electrical Engineering Fundamentals**, Prentice Hall of India.
3. Kothari D. P., Nagrath I. J., **Theory and Problems of Basic Electrical Engineering**, Prentice Hall of India.
4. Hayt W. H., Kemmerly J. E. and Durbin S. M., **Engineering Circuit Analysis**, Tata McGraw-Hill Publishing Company Limited.
5. Chapman S. J., **Electric Machinery Fundamentals**, McGraw Hill Book Company.
6. Hughes E., **Electrical & Electronic Technology**, 8th Edition, Pearson Publishing